

ASSEMBLE AND FLY DRONE – UAV BOOTCAMP

The ECE Society, BIT Sindri, successfully organized a five-day “Assemble and Fly Drone” Bootcamp from December 3 to December 7, 2024. The event provided students with practical exposure to assembling, understanding, and flying unmanned aerial vehicles (UAVs), along with insights into sensors, electronics, and communication systems. The bootcamp was conducted under the guidance of the ECE Department faculty.

Day 1 – December 3, 2024

The bootcamp began with an inaugural session by Dr. Pankaj Rai, Director, BIT Sindri. A team of experts from IIT (ISM) Dhanbad—including Dr. Varun Choudhary, Mr. Aksh Raj, Mr. Kundan Kumar Suman, Prof. Ravi Kumar Gangwar, Prof. J. Singh T., and Prof. Samrat Mukhopadhyay—introduced participants to UAV fundamentals, drone components, and system architecture. Nearly 60 students took part, showcasing great enthusiasm toward drone technology.

Day 2 to Day 4 – Technical & Practical Training:

The following days covered:

- Drone assembly and component alignment.
- Communication protocols and embedded systems.
- Sensor integration and flight-control electronics.
- Basics of aerodynamics and flight stability.

Students engaged in both theoretical learning and extensive hands-on sessions, assembling drone modules step-by-step under expert guidance.

Final Day – December 7, 2024

The concluding day featured a session on UAV control systems by Dr. Varun Choudhary, followed by an MCQ-based assessment. The bootcamp concluded with a flight-testing session where participants successfully flew the drones they assembled. A certificate ceremony was held in the presence of the Director and ECE faculty members, appreciating the efforts of all experts and participants.

